

Kshitiz Bhurtel

+977 9806766171 • kshitizbhurtel@gmail.com • [LinkedIn](#)

Experienced Software Engineer with over 3+ years of expertise in full-stack development. Skilled in React, Django, and, Spring Boot, with a focus on optimizing system performance and enhancing data security. Proven ability to lead teams, develop scalable applications, and implement practical solutions to improve efficiency and reduce operational costs. Experienced with AWS, Jenkins, and OpenCV for real-time data processing. Focused on delivering impactful results through data-driven development and practical technology solutions.

Professional Experience

SOFTWARE ENGINEER II | DELOITTE USI | MAR 2022 – MAY 2025

Backend Lead, Cybersecurity Benchmarking

- Developed a **cybersecurity benchmarking application** used by 40+ organizations to assess maturity levels, identify threats, analyze gaps, and recommend risk mitigation measures, **improving security posture** by **27%**.
- Designed and implemented **authorization, RBAC, and benchmarking logic** using Django and Python, ensuring **~100%** secure access control and enhancing reporting accuracy by **35%**.
- **Led a team of 3 backend developers**, conducting weekly code reviews, which **improved code quality**.
- Optimized application scalability and performance by deploying on AWS and integrating with PostgreSQL, achieving a **15% reduction in response time**.

Software Engineer, Autonomous Driving Synthetic Data generation and testing

- Imported custom vehicles (e.g., firetruck, ambulance, police car, SUVs) into **CARLA**, defining unique classes for each vehicle type to enhance simulation diversity.
- Simulated real-time crash scenarios in Carla Simulation using Python scripts and tested them against the Autoware AD Stack, identifying 12% more edge cases and improving safety validation efficiency by 5%.
- Generated **synthetic dataset for AD models**, creating **100K+** images with semantic and instance segmentation, LIDAR, RADAR reducing data collection costs by **~80%** compared to real-world data acquisition.
- Reduced processing time for **2D bounding box generation** from **20 seconds to 1 second** while ensuring **accurate object detection and tracking** through **instance and semantic segmentation**.
- Introduced various types of **image noise** (salt and pepper, Gaussian, water droplets, film grain) to **enhance dataset diversity** and **improve model robustness**, resulting in a **5% increase in model accuracy** during validation on diverse test scenarios.
- Trained models using the synthetic dataset, achieving **5%** improvement in accuracy compared to real-world data, demonstrating the effectiveness of simulated training data for diverse driving conditions.
- Streamlined dataset generation pipeline using **Jenkins** and **AWS**, reducing manual intervention by **75%**.

Contract Experience

FULL STACK DEVELOPER | BITPOSTAGE | OCT 2023 – PRESENT

- Designed and developed a **crypto-powered postage purchasing platform**, enabling users to seamlessly buy postage labels using cryptocurrencies from **USPS, UPS and FEDEX**

- The application was built in React frontend and **FastAPI backend**, deployed on **AWS** to ensure **99.9% uptime** and **scalable infrastructure** for future growth.
- **Replaced a third-party payment verification system** with a **custom-built BTC, ETH, XMR, and XRP validator**, **reducing transaction verification time** by **60%** and **cutting third-party costs** by **100%**.

Skills

Java • Python • JavaScript • React • Spring Boot • Django • REST APIs • PostgreSQL • MongoDB • DB2 • AWS (S3, EC2, Lambda, RDS, EKS) • Jenkins • WebSockets • OpenCV • Carla Simulation • Autoware • Microservices • Cloud Computing • Computer Vision • Payment Verification System • Software Optimization • Agile Development • Technical Leadership • Code Reviews • Performance Tuning

Education

AUG 2018 – AUG 2022

BE in Computer Science | Visvesvaraya Technological University | Bangalore, India

Activities

Reading • Doodling • Exploring new tech trends